

Mathematical Analysis

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Increasing and decreasing functions

Definition

We will say that f is an increasing function at point y if there exists $\delta > 0$ such that $f(x) \leq f(y)$ for $x \in (y - \delta, y)$ and $f(x) \geq f(y)$ for $x \in (y, y + \delta)$.

Remark

If f is an increasing function for any $y \in [a, b]$, then $f(a) \leq f(b)$.

Similar definitions for decreasing functions and strictly increasing functions.

Theorem

If f is smooth and $f'(y) > 0$ for any $y \in [a, b]$, then f is strictly increasing.

Theorem

If f is an increasing function and there exists $f'(y)$, then $f'(y) \geq 0$.

Example

$f(x) = x^3$ – strictly increasing function but $f'(0) = 0$.

Definition

We will say that f has local maximum at point x if for some $\delta > 0$, $f(x) \geq f(y)$ for any $y \in (x - \delta, x + \delta)$.

Theorem

If f has local maximum at x and there exists $f'(x)$, then $f'(x) = 0$.

Theorem (Rolle's theorem)

If f is smooth on $[a, b]$ and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.

Cauchy theorem

Theorem

If f, g are continuous on $[a, b]$, smooth on (a, b) and $g' \neq 0$, then there exists $c \in (a, b)$ such that

$$\frac{f(a) - f(b)}{g(a) - g(b)} = \frac{f'(c)}{g'(c)}.$$

Theorem (Lagrange theorem)

If f is continuous on $[a, b]$ and smooth on (a, b) , then there exists $c \in (a, b)$ such that

$$\frac{f(a) - f(b)}{a - b} = f'(c).$$

Theorem

Let f be a smooth function on (a, b) .

- 1. If $f' \geq 0$, then f is an increasing function;*
- 2. If f is an increasing function, then $f' \geq 0$;*
- 3. If $f' > 0$, then f is strictly increasing function.*

Theorem

Let f be a smooth function on $[a, b]$. If $f'(x) = 0$ for any $x \in (a, b)$, then $f \equiv \text{const}$.

Theorem

If f is smooth on (a, b) and $f' \neq 0$, then $f' > 0$ or $f' < 0$.

Theorem

If f is smooth on (a, b) and $f'(u) < \alpha < f'(v)$, $u < v$, then there exists $c \in (u, v)$ such that $f'(c) = \alpha$.

f' is not continuous!

L'hopitals rule

Theorem

Let f be a continuous function on $[a, b)$, smooth on (a, b) and there exists $\lim_{x \rightarrow a+} f'(x) = d$. Then there exists $f'(a)$ and $f'(a) = d$.

Theorem

Let $\lim_{x \rightarrow a+} f(x) = \lim_{x \rightarrow a+} g(x) = 0$. If f and g is smooth and $g' \neq 0$ on (a, b) , then

$$\lim_{x \rightarrow a+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a+} \frac{f'(x)}{g'(x)}.$$

Same for $\pm\infty$.

Definition

Function f'' is a derivative of f' , $f^{(n+1)} := (f^{(n)})'$.

Example

Polynomial with prescribed derivatives $P^{(n)}(a) = d_n$,

$$P(x) = d_0 + \frac{d_1}{1!}(x-a) + \frac{d_2}{2!}(x-a)^2 + \dots + \frac{d_n}{n!}(x-a)^n.$$

Existence and uniqueness.

Taylor polynomial

Definition

We will say that $P(x)$ is a Taylor polynomial of degree n if

$$f(x) - P(x) = o((x - a)^n), \quad x \rightarrow a.$$

Theorem

If P_1 and P_2 Taylor polynomials of degree n , then $P_1 = P_2$.

Taylor formula (Peano)

Theorem

Let $f : (a, b) \rightarrow \mathbb{R}$ and there exist $f', \dots, f^{(n-1)}$ on $(t - \delta, t + \delta)$ for some $\delta > 0$ and there exist $f^{(n)}(t)$. Then

$$\begin{aligned} f(x) = f(t) + \frac{f'(t)}{1!}(x - t) + \frac{f''(t)}{2!}(x - t)^2 + \dots \\ \dots + \frac{f^{(n)}(t)}{n!}(x - t)^n + o((x - t)^n). \end{aligned}$$

Lemma

If there exist $g', \dots, g^{(n-1)}$ on $(t - \delta, t + \delta)$ for some $\delta > 0$ and there exist $g^{(n)}(t)$ and $g(t) = g'(t) = g''(t) = \dots = g^{(n)}(t) = 0$, then $g(t) = o((x - t)^n)$.

Theorem

Let $f(t) = f'(t) = \dots = f^{(n)}(t) = 0$ and $f^{(n+1)}(t) \neq 0$. If n is even, then t is not an extreme point. If n is odd, then t is an extreme point.

This is very general scheme.

Taylor formula (Lagrange)

Theorem

Let $f : (a, b) \rightarrow \mathbb{R}$ and there exist $f', \dots, f^{(n+1)}$ on $(t - \delta, t + \delta)$ for some $\delta > 0$. Then there exists $\xi \in (x, t)$ such that

$$f(x) = f(t) + \frac{f'(t)}{1!}(x - t) + \frac{f''(t)}{2!}(x - t)^2 + \dots \\ \dots + \frac{f^n(t)}{n!}(x - t)^n + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - t)^{n+1}.$$

Lemma

Case $f(t) = f'(t) = f''(t) = \dots = f^{(n)}(t) = 0$.